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Dynamic response and resilience of unmanned aerial vehicle swarms under electromagnetic interference

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ABSTRACT

Electromagnetic interference (EMI) is among the most effective means of disrupting UAV swarms. However, there remains a lack of quantitative models that characterize its impact on swarm coordination. To bridge this gap, we propose a modeling framework that quantifies EMI-induced degradation in collective swarm performance. The framework first evaluates EMI effects on navigation and communication systems, then incorporates the resulting errors in state estimation and command transmission into a swarm dynamics model to assess their cumulative impact on the coordination of UAV swarms. To enable systematic evaluation, we further define three indices, capturing formation distortion, trajectory deviation, and communication connectivity, that jointly quantify swarm-level coordination degradation under EMI. Numerical experiments show that as the jamming power increases, UAV coordination degrades significantly. For a fixed total jamming power, the more spatially distributed the jamming sources are, the more severely the swarm's trajectory is deviated and its communication connectivity is degraded. Moreover, the widely adopted V-shaped formation demonstrates the lowest resilience among the four tested UAV swarm formations under EMI.

1. Introduction

The advancement of unmanned aerial vehicle (UAV) swarms has enabled transformative applications across diverse domains, including precision agriculture, military operations, cargo transportation, and surveillance [1]. Effective coordination and deployment of such swarms depend critically on the reliable operation of wireless communication and navigation systems [2,3]. Specifically, global navigation satellite systems (GNSS) provide the primary source of positioning and location awareness [4]. Inter-UAV coordination in large-scale swarms relies on the timely exchange of neighbor state information and high-level commands, such as desired trajectories or formation references, over wireless links [5,6]. In this context, electromagnetic interference (EMI) has emerged as a critical threat to UAV swarm operations, owing to its potent ability to disrupt both communication networks and GNSS-based navigation, thereby compromising swarm coherence, safety, and mission effectiveness [7,8].

A substantial body of research has investigated the resilience of UAV swarms under various adversarial conditions [9–12]. Some studies adopt concepts from complex systems theory, such as percolation theory [9], to characterize swarm robustness from a network-theoretic perspective. Others focus on mission-oriented performance. Feng *et al.* [11] study the relationship between swarm size and mission reliability, introducing importance measures and proposing an optimization framework for allocating UAV types to maximize reliability. Moving beyond static resilience metrics, Zhang *et al.* [10] introduce a dynamic evaluation approach tailored to cross-domain swarms operating in uncertain adversarial environments. Xu *et al.* [13] propose a method for collision avoidance and interference mitigation in UAV swarms where multiple UAVs cooperatively track a common target.

Compared with other adversarial conditions, EMI represents a fundamentally different threat modality [14,15]. As an emerging and non-kinetic counter-UAV measure, EMI does not result in the permanent loss of UAVs or direct node failures [16]. Instead, it disrupts the wireless communication links by degrading the signal-to-interference-plus-noise ratio across the network, leading to packet loss, reduced transmission range, and intermittent connectivity. This results in a progressive or localized breakdown of information exchange, although all nodes remain physically operational. Consequently, the resilience of UAV swarms under EMI cannot be adequately assessed using traditional fault-tolerance models based on node removal alone, necessitating new analytical frameworks that account for partial and dynamic link degradation.

Maintaining communication connectivity in the presence of node failures, whether due to physical damage or cyber attacks, has received considerable attention [17]. These studies often focus

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on topology reconfiguration, fault-tolerant routing, and trajectory planning to maintain communication connectivity when certain agents are removed from the network. UAV swarms are closely related to wireless sensor networks. Ensuring robust k-connectivity is critical for network reliability [18]. Recent work introduces a concept of security entropy and uses this as a foundation measure to improve communication connectivity of the UAV swarm [19]. In the domain of topology reconfiguration, recent work by Liu *et al.* [20] introduce the concept of a Multi-hop Differential Sub-Graph (MDSG) to represent local damage and facilitate efficient recovery trajectory control for the remaining UAVs. Li *et al.* [21] consider the impact of jamming on UAV swarm communication and propose a topology optimization method in which each UAV transmits data to a collection center either directly or via relay through other UAVs, while accounting for malicious jamming. Hu *et al.* [22] incorporate the consensus speed while constructing communication topologies, and propose a dynamic recovery method is proposed under attacks by establishing the relationship between algebraic connectivity and consensus speed. Furthermore, since the performance of communication links in UAV swarms is highly dependent on the environment, trajectory design plays a crucial role in enhancing communication performance. Xing *et al.* [23] propose a hierarchical control strategy to address the challenge of communication disruption caused by EMI. Wang *et al.* [24] formulate an optimization problem that jointly maximizes throughput and minimizes UAV energy consumption, taking into account interference among coexisting users.

However, a general modeling framework to quantify how EMI affects the coordination of UAV swarms has not yet been established. Such a framework would support the development of more resilient UAV swarms and could also inform the deployment of EMI-based defensive systems for protecting critical infrastructure against the attacks of UAV swarms. By characterizing how EMI disrupts key swarm capabilities, such as formation geometry, inter-vehicle communication [25], and trajectory tracking, a modeling framework enables UAVs to proactively reconfigure formations [26], adapt communication topologies [27], and adjust flight paths to mitigate or avoid interference. From the standpoint of electronic defense, the impact of EMI on UAV coordination can also facilitate the systematic optimization of jamming strategies, particularly with respect to timing, spatial targeting, and sequencing, thereby maximizing disruptive effectiveness while accounting for resource constraints.

UAV swarms are capable of executing a wide range of missions, many of which rely on coordinated behaviors such as following preplanned trajectories or maintaining specific formations. For instance, in surveillance missions, UAV trajectories must be carefully designed to ensure full coverage of target areas. This paper proposes a modeling framework that quantifies the impact of EMI on UAV swarm coordination across diverse operational scenarios. The contributions of this work are listed as follows:

1. A Control-Agnostic and Physics-Informed Interference Modeling Framework: We propose a framework that abstracts the impact of EMI on UAV swarm performance into two universal interfaces: (i) GNSS-derived state requirements and (ii) inter-agent communication dependencies.
2. A Set of Task-Performance-Correlated yet Scenario-Generalizable Indicators: We propose three performance indices that cover both control performance and communication quality aspects. These indices directly reflects mission effectiveness while remaining applicable across diverse task scenarios without scenario-specific recalibration.
3. A Dual-Perspective Strategy Evaluation Framework. The proposed modeling framework enables quantitative assessment of offensive interference deployment and defensive swarm strategies, supporting data-driven optimization for both adversarial and resilient operational planning.

The remainder of this paper is organized as follows. Section 2 presents the modeling framework for UAV swarms, encompassing system dynamics and communication networks. Section 3 investigates the influence of EMI on navigation systems and communication networks of UAV swarms. Section 4 list three indicators quantify the influence of EMI on swarm-level coordination, capturing formation distortion, trajectory deviation, and communication connectivity. Section 5 reports numerical experiments numerical experiments across multiple scenarios to evaluate how key deployment strategies of both EMI systems and UAV swarms that affect jamming performance. Finally, Section 6 concludes the paper and suggests potential directions for future research.

2. Problem description and system model

EMI degrades a UAV node's wireless communication capability. As a result, both the estimation of its own state using GNSS and the reception of control or positional data from other swarm members via intra-swarm communication become unreliable. Since the movement of each UAV is determined by a feedback loop involving self-state estimation and relative information from neighbors, these EMI-induced errors propagate into the control inputs, leading to significant deviations in trajectory and formation integrity. The process is illustrated in Fig. 1.

In this section, we demonstrate how swarm states, and how inter-agent communication patterns couple node dynamics. As a concrete example, we employ distributed leader-follower control to illustrate the framework's application. Crucially, the proposed modeling framework in Section 3 enables integration with diverse control paradigms, as long as the GNSS-derived state requirements and inter-agent communication dependencies are specified.

2.1. General modeling framework of swarm dynamics

Command dissemination and state estimation are fundamental to effective UAV swarm control. In this work, we use a distributed leader–follower architecture with multiple leaders as an example, a paradigm widely employed in real-world engineering systems. By varying the number of leaders and the number of followers assigned to each leader, the control structure can be flexibly tuned between more distributed and more centralized configurations. Consequently, the insights gained from the proposed modeling framework are generalizable across a range of control architectures, such as relative measurement-based control [28].

Consider a UAV swarm comprising N nodes. For UAV swarm motion, the fundamental state vector comprises position and velocity. The dynamics of each node $i \in \{1, \dots, N\}$ can be modeled as a double integrator [29]. The state-space representation is given by

$$\begin{aligned} \dot{\mathbf{p}}_i &= \mathbf{v}_i \\ \dot{\mathbf{v}}_i &= \mathbf{u}_i \end{aligned}, \tag{1}$$

where $\mathbf{p}_i \in \mathbb{R}^3$ and $\mathbf{v}_i \in \mathbb{R}^3$ denote the position and velocity vectors of node i in the global coordinate frame, respectively.

The desired states $\mathbf{p}_i^d$ and $\mathbf{v}_i^d$ are generated by a high-level coordination policy, $\mathcal{P}$, which encapsulates the specific swarm control objective. The UAVs are partitioned into K disjoint groups $\mathcal{V}_1, \dots, \mathcal{V}_K$, where $N_k = |\mathcal{V}_k|$ denotes the number of UAVs in the k^{th} group. Let r_k denote the reference node of group $\mathcal{V}_k$, which provides a reference position for controlling the UAVs in $\mathcal{V}_k$. The desired states for nodes i in $\mathcal{V}_k$ are defined as

$$\begin{aligned} \mathbf{p}_i^d &= \mathbf{p}_{r_k} + \mathbf{d}_i^f \\ \mathbf{v}_i^d &= \mathbf{v}_{r_k} \end{aligned}, \tag{2}$$

where $\mathbf{d}_i^f \in \mathbb{R}^3$ is the desired relative offset of node i from its reference node r_k. The control input of the reference node $\mathbf{u}_{r_k}$ is generated independently to track a global reference trajectory $(\mathbf{p}_{r_k}^{ref}, \mathbf{v}_{r_k}^{ref})$.

The control input $\mathbf{u}_i \in \mathbb{R}^3$ represents the acceleration command. Control laws, as embodied by the control input $\mathbf{u}_i$, enables cohesive swarm

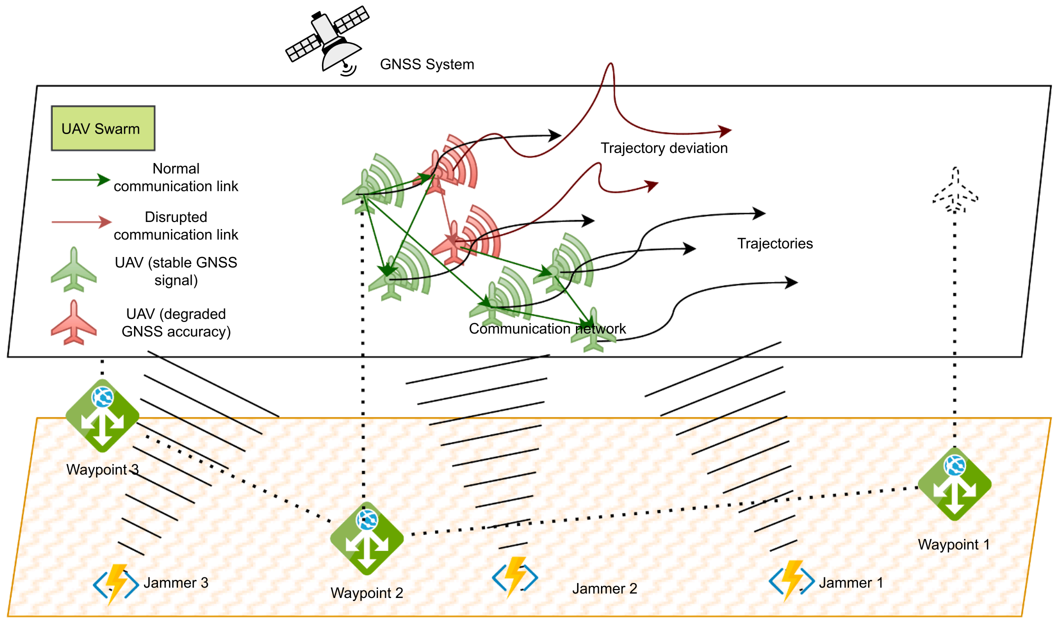

Fig. 1. Illustration of the impact of distributed EMI jammers on UAV swarm coordination while navigating through three predefined waypoints.

mobility under various coordination strategies. To illustrate the mechanism by which individual UAVs are driven toward their desired states, we adopt a general proportional-derivative (PD) control law, expressed as

$$\mathbf{u}_i = \mathbf{K}_p(\mathbf{p}_i^d - \mathbf{p}_i) + \mathbf{K}_d(\mathbf{v}_i^d - \mathbf{v}_i), \tag{3}$$

where $\mathbf{p}_i^d$ and $\mathbf{v}_i^d$ are the *desired position* and *desired velocity* for agent i, respectively. Here, $\mathbf{K}_p = k_p\mathbf{I}_3$ and $\mathbf{K}_d = k_d\mathbf{I}_3$ are positive definite diagonal gain matrices ensuring system stability, where $\mathbf{I}_3$ represents an identity matrix with the dimension of three. In practice, physical UAV actuators impose limits on achievable acceleration. To reflect realistic dynamics, the control command is saturated such that $\|\mathbf{u}_i\| \leq u_{\max}$, where $u_{\max}$ is set to $8.0\ \mathrm{m/s^2}$ in the following of this paper, a conservative yet representative bound for small-to-medium multirotor platforms used in swarm missions.

The control input for guiding a UAV swarm along a desired trajectory is primarily determined by two key factors. First, the set of target waypoints that the swarm aims to reach, which defines the overall mission path. Second, the desired formation geometry, which governs the spatial configuration of individual UAVs and enables role-specific functionalities such as coverage, sensing, or communication relay. We assume that the entire swarm follows a common reference trajectory while maintaining specified relative positions. Let $\mathbf{p}_{r_k}(t)$ be the position of the reference node r_k, and let $\boldsymbol{\delta}_i(t)$ denote the time-varying desired offset of UAV i relative to the reference node. The reconfiguration process is scheduled over a finite time horizon $[0, T_f]$, where $T_f > 0$ represents the total duration required to transition from the initial formation to the target formation. The desired offset evolves according to

$$\boldsymbol{\delta}_i(t) = \begin{cases} \boldsymbol{\delta}_i^{\text{init}}, & t = 0, \\ \boldsymbol{\delta}_i^{\text{target}}, & t = T_f, \end{cases} \tag{4}$$

with smooth interpolation for $t \in (0, T_f)$, ensuring continuous and collision-free trajectory transitions. The desired trajectory for UAV i is defined as

$$\mathbf{p}_i^{\text{des}}(t) = \mathbf{p}_{\text{ref}}(t) + \boldsymbol{\delta}_i(t), \tag{5}$$

where $\boldsymbol{\delta}_i(t)$ is parameterized using a cubic polynomial, given by

$$\boldsymbol{\delta}_i(t) = \boldsymbol{\delta}_i^{\text{init}} + \left(\boldsymbol{\delta}_i^{\text{target}} - \boldsymbol{\delta}_i^{\text{init}}\right) \cdot w(t), \tag{6}$$

and $w(t) = 3(t/T_f)^2 - 2(t/T_f)^3$ ensures zero initial and final acceleration.

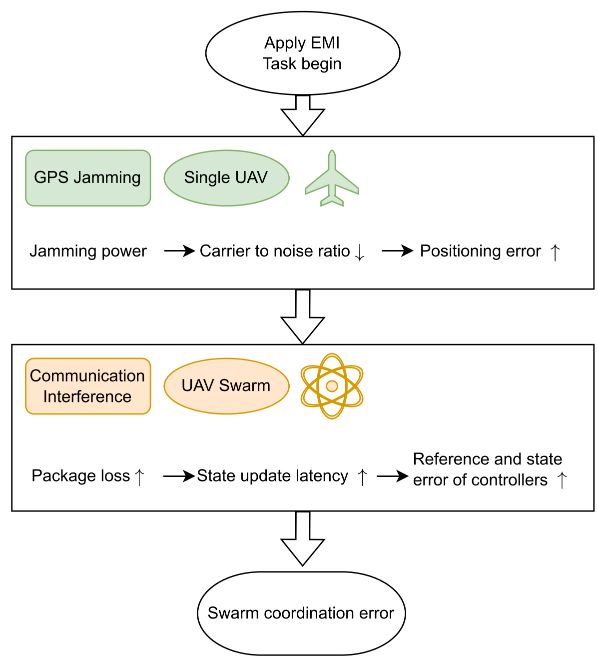

Fig. 2. Flowchart illustrating how EMI degrades UAV swarm coordination. EMI first increases the positioning error of individual UAVs; this error then propagates across the swarm. Additionally, EMI jams communication links, causing packet loss that further amplifies reference and state estimation errors in the control loops, ultimately leading to coordination error of UAV swarms.

2.2. *Impact of jamming attacks on swarm dynamics*

For GNSS, EMI introduces errors in a node's self-localization. We assume that the formation is maintained based on the relative distance of each follower node to a reference node. When the affected node is a reference node, its erroneous position estimate propagates to all follower nodes in its groups, amplifying the overall impact across the swarm. The mechanism is described in Fig. 2. Let $\mathbf{p}_i$ denote the true position of node i, and $\hat{\mathbf{p}}_i$ its GNSS-derived estimate. The desired position $\mathbf{p}_i^{\text{des}}$ for follower i is derived based on the state received from its leader. If the communication channel is jammed, delays or packet losses may occur, leading to outdated or corrupted state information and thus errors in $\mathbf{p}_i^{\text{des}}$. The actual desired position influenced by GNSS becomes

$$\hat{\mathbf{p}}_i^{\text{des}} = \mathbf{p}_i^{\text{des}} + \Delta\mathbf{p}_i^{\text{des}}, \tag{7}$$

where $\Delta\mathbf{p}_i^{\text{des}}$ represents the cumulative estimation error due to EMI in both sensing and communication subsystems. This compound error constitutes the primary modeling focus of the following sections.

2.3. Communication network of UAV swarms

We denote the set of all UAVs in the swarm as $\mathcal{V} = \{1, 2, \dots, n\}$, where each node $i \in \mathcal{V}$ represents an individual agent equipped with sensing, computation, and wireless communication capabilities. These agents collectively form a dynamic communication network, modeled as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the vertex set (i.e., the UAVs), and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set representing active communication links between pairs of UAVs [6]. A link $(i, j) \in \mathcal{E}$ exists if the signal-to-interference-plus-noise ratio (SINR) from node j to node i exceeds a predefined threshold required for reliable packet reception, which depends on modulation, coding, and environmental conditions.

We define the initial communication topology as $\mathcal{G}^{\text{init}} = (\mathcal{V}, \mathcal{E}^{\text{init}})$, representing the network structure in the absence of interference. Under EMI, the topology evolves such that communication links may be dynamically removed due to link failures, but no new links are formed. That is, $\mathcal{E}(t) \subseteq \mathcal{E}^{\text{init}}$ for all $t \geq 0$, reflecting a conservative assumption of non-enhancing connectivity under attack. The network $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$ is spatially embedded and time-varying, with its evolution governed by UAV mobility and the spatial distribution of interference. The loss of edges can lead to network partitioning, impairing distributed coordination and consensus-based control.

3. Modeling the influence of electromagnetic jamming

Multiple jamming techniques are employed in EMI systems, including single-frequency (continuous-wave) jamming, pulse jamming, and sweep jamming, each characterized by distinct temporal, spectral, and power profiles. These modalities differ in terms of carrier frequency, pulse duration, repetition rate, sweep range, and transmitted power, leading to varying degrees of disruption on communication links within UAV swarms. To enable a unified and analytically tractable representation, we model the jamming effect as a time-varying interference power signal.

3.1. Spatial modeling of jamming power

The jammer equipments employ high-power signals to disrupt the physical layer of wireless communications and GNSS by reducing the SINR below the receiver's operational threshold. In this work, we assume that jammers target the GNSS and communication networks of UAV swarms independently. This subsection provides a unified modeling framework for the jamming signals targeting UAV swarms. However, the specific parameter in this framework must be set differently for GNSS and communication interference scenarios. We denote the set of jammer as $\mathcal{V}^{\text{jam}}$. We consider a scenario in which the jammers are placed in three-dimensional space near the target area and transmits their jamming signal $P_{h\in\mathcal{V}^{\text{jam}}}^{\text{jam}}$ toward ground-based base stations. During jamming attacks, the effective received power at a node is significantly influenced by the jammer's transmission power and relative position, as characterized by Eq. (14). This spatially heterogeneous interference leads to non-uniform performance degradation across the network, resulting in disconnected components and compromising swarm coherence.

The jamming-to-signal ratio (JSR) is defined in the linear domain as

$$\text{JSR} = \frac{P_{ih}^{\text{jam, lin}}}{P_i^{\text{rec, lin}}}, \tag{8}$$

where $P_{ih}^{\text{jam, lin}}$ denotes the received jamming power from jammer h at victim node i (in W), and $P_i^{\text{rec, lin}}$ represents the desired signal power received from the legitimate transmitter (in W). Both quantities are primarily governed by the Euclidean distance between the respective transmitter and the receiver [30]. The Euclidean distance between any node i and jammer h is given by

$$d_{ih}^{\text{jam}} = \|\mathbf{p}_i - \mathbf{p}_h\|, \quad i \in \mathcal{V}, \quad h \in \mathcal{V}^{\text{jam}}, \tag{9}$$

where $\mathbf{p}_i$ and $\mathbf{p}_h$ represent the 3D coordinates of node i and jammer h, respectively (in m). Under free-space propagation conditions, the received jamming power in linear scale is modeled by the Friis transmission formula, given by

$$P_{ih}^{\text{jam, lin}} = P_h^{\text{jam, lin}} G_h G_i \left(\frac{\lambda}{4\pi d_{ih}^{\text{jam}}} \right)^2, \tag{10}$$

where $P_h^{\text{jam, lin}}$ is the jammer's transmit power (in W), G_h and G_i are the antenna gains of the jammer and receiver, respectively, and λ is the carrier wavelength (in m). Since GNSS signals and the communication networks of UAV swarms operate over distinct bandwidths, effective jamming requires bandwidths tailored to each interference objective. Accordingly, the parameter λ is set differently for GNSS jamming and communication network interference. For analytical convenience and consistency with conventional practices in wireless communications, power quantities are often expressed in logarithmic units. The corresponding received jamming power is given by

$$P_{ih,\text{dBW}}^{\text{jam}} = P_{h,\text{dBW}}^{\text{jam}} + G_{h,\text{dBi}} + G_{i,\text{dBi}} - 20\log_{10}\left(\frac{4\pi d_{ih}^{\text{jam}}}{\lambda} \right), \tag{11}$$

where $P_{h,\text{dBW}}^{\text{jam}}$ denotes the jammer's transmit power in dBW.

The free-space path loss at a reference distance d_0 (typically $d_0 =$ 1 m) is defined as

$$L_0 = 20\log_{10}\left(\frac{4\pi d_0}{\lambda} \right), \tag{12}$$

Consequently,

$$20\log_{10}\left(\frac{\lambda}{4\pi d_{ih}^{\text{jam}}} \right) = -L_0 - 20\log_{10}\left(\frac{d_{ih}^{\text{jam}}}{d_0} \right). \tag{13}$$

Substituting into Eq. (11), the received jamming power can be reformulated as

$$P_{ih,\text{dBW}}^{\text{jam}} = P_{h,\text{dBW}}^{\text{jam}} + G_{h,\text{dBi}} + G_{i,\text{dBi}} - L_0 - 20\log_{10}\left(\frac{d_{ih}^{\text{jam}}}{d_0} \right) \tag{14}$$

To distinguish between the two interference scenarios, we adopt the superscripts "GNSS" and "COMM" for parameters associated with GNSS and UAV swarm communication networks, respectively. For instance, the jammer' s transmit power is denoted as $P_h^{\text{jam, GNSS}}$ for GNSS jamming and $P_h^{\text{jam, COMM}}$ for communication network jamming.

3.2. Modeling of navigation interference: GNSS degradation

Global Navigation Satellite System (GNSS) receivers operating in the L1 band (1575.42 MHz) exhibit high susceptibility to electromagnetic interference. This section models the propagation of jamming effects from power degradation to positioning error, based on carrier-to-noise ratio collapse and geometric dilution of precision (GDOP) principles [31].

The carrier-to-noise-density-plus-interference ratio at the GNSS receiver input of UAV i under interference from jammer h is given by [32,33]

$$\frac{C^{\text{GNSS}}}{N_0^{\text{GNSS}} + J_{ih}^{\text{GNSS}}(t)} = \frac{C^{\text{GNSS}}}{N_0^{\text{GNSS}} + P_{ih}^{\text{jam, GNSS}}(t)/B_h^{\text{GNSS}}}, \tag{15}$$

where C^{GNSS} denotes the received GNSS signal power (in W), N_0^{GNSS} is the thermal noise power spectral density over the GNSS bandwidth (in W/Hz), $P_{ih}^{\mathrm{jam,GNSS}}(t)$ represents the jamming power received from jammer h targeting the GNSS of UAV i (in W), and B_h^{GNSS} is the jammer bandwidth (in Hz) targeting the GNSS of UAV swarms. The jamming power spectral density is defined as $J_{ih}^{\mathrm{GNSS}}(t) = P_{ih}^{\mathrm{jam,GNSS}}(t)/B_h^{\mathrm{GNSS}}$, with units of W/Hz. The ratio in logarithmic units is expressed as

$$\left(\frac{C^{\mathrm{GNSS}}}{N_0^{\mathrm{GNSS}} + J_{ih}^{\mathrm{GNSS}}(t)}\right)_{\text{dB-Hz}} = C_{\mathrm{dBW}}^{\mathrm{GNSS}} - \left(N_0^{\mathrm{GNSS}} + J_{ih}^{\mathrm{GNSS}}(t)\right)_{\mathrm{dBW/Hz}}, \tag{16}$$

where the combined noise-plus-jamming spectral density in dBW/Hz is computed from linear quantities as

$$\left(N_0^{\mathrm{GNSS}} + J_{ih}^{\mathrm{GNSS}}(t)\right) = 10\log_{10}\left(10^{(N_0)/10} + 10^{\left(J_{ih}^{\mathrm{GNSS}}(t)\right)/10}\right), \tag{17}$$

with the conversions from linear to logarithmic scale given by

$$\begin{aligned} C_{\mathrm{dBW}}^{\mathrm{GNSS}} &= 10\log_{10}\left(\frac{C^{\mathrm{GNSS}}}{1\ \mathrm{W}}\right), \\ N_{0,\mathrm{dBW/Hz}}^{\mathrm{GNSS}} &= 10\log_{10}\left(\frac{N_0^{\mathrm{GNSS}}}{1\ \mathrm{W/Hz}}\right), \\ J_{ih,\mathrm{dBW/Hz}}^{\mathrm{GNSS}}(t) &= 10\log_{10}\left(\frac{J_{ih}^{\mathrm{GNSS}}(t)}{1\ \mathrm{W/Hz}}\right). \end{aligned} \tag{18}$$

The positioning error standard deviation is determined by both pseudorange measurement accuracy and satellite geometry, derived by

$$\sigma_{\mathrm{pos}}(t) = \mathrm{GDOP} \cdot \sigma_{\mathrm{pr}}(t), \tag{19}$$

where GDOP represents the geometric dilution of precision (a dimensionless factor reflecting satellite constellation geometry), and $\sigma_{\mathrm{pr}}(t)$ denotes the standard deviation of pseudorange measurements. The pseudorange error exhibits an inverse square root relationship with carrier-to-noise-density-plus-interference ratio, given by

$$\sigma_{\mathrm{pr}}(t) \propto \frac{1}{\sqrt{C^{\mathrm{GNSS}}/(N_0^{\mathrm{GNSS}} + J_{ih}^{\mathrm{GNSS}}(t))}}. \tag{20}$$

Thus, jamming-induced reduction in $C^{\mathrm{GNSS}}/(N_0^{\mathrm{GNSS}} + J_{ih}^{\mathrm{GNSS}}(t))$ increases pseudorange error $\sigma_{\mathrm{pr}}(t)$, which in turn amplifies the positioning error $\sigma_{\mathrm{pos}}(t)$, especially under unfavorable geometric conditions characterized by high GDOP values, such as in urban canyons or during directional jamming scenarios. Let $\hat{\mathbf{p}}_1(t)$ be the GNSS-reported leader position. In practice, due to the iterative nature of position determination, the widespread use of Kalman filtering in receivers, and temporal correlations in measurement errors, the estimation error does not always follow a zero-mean normal distribution. Instead, empirical studies have shown that positioning errors often exhibit systematic biases [34]. Therefore, to account for potential deterministic deviations, we introduce a bias term $\mathbf{b}(t)$ into the error model. The estimation error is modeled as

$$\Delta\mathbf{p}_i^{\mathrm{GNSS}}(t) = \hat{\mathbf{p}}_i(t) - \mathbf{p}_i(t) \sim \mathcal{N}\left(\mathbf{b}(t),\ \sigma_{\mathrm{pos}}^2(P_{ih}^{\mathrm{jam,\ GNSS}}(t)) \cdot \mathbf{I}_3\right), \tag{21}$$

where $\mathbf{I}_3$ denotes the 3×3 identity matrix, and $\sigma_{\mathrm{pos}}(P_{ih}^{\mathrm{jam,\ GNSS}}(t))$ is dynamically determined by current received jamming power $P_{ih}^{\mathrm{jam,\ GNSS}}(t)$.

3.3. Modeling of communication interference

Under the modeling framework described in Section 2.1, state information originates from designated reference nodes and propagates through the swarm via a multi-hop broadcast protocol. At each time step, the reference nodes $\{r_k\}_{k=1}^K$ update their local memory with their GNSS-derived position estimates $\hat{\mathbf{p}}_{r_k}$. Nodes within each group of the swarm then attempt to acquire the state of their assigned reference node through direct or indirect communication links. Specifically, node i in group k first checks whether the communication link to its assigned reference node r_k is active. If the link is active, node i directly adopts $\hat{\mathbf{p}}_{r_k}$. Otherwise, it searches among its one-hop neighbors that possesses a valid estimate of $\hat{\mathbf{p}}_{r_k}$. Upon finding such a relay, node i accepts the relay estimate. If no valid relay node is available, node i retains the estimate of $\hat{\mathbf{p}}_{r_k}$ from the previous time step as a short-term holdover. Let $\mathbf{s}_j(t) = [\mathbf{p}_j(t), \mathbf{v}_j(t)]$ denote the state of UAV j. Due to EMI, UAV i receives a corrupted and delayed version:

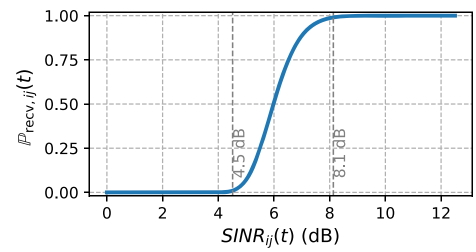

Fig. 3. The "cliff" behavior of the relationship between $\mathbb{P}_{\mathrm{recv},ij}(t)$ and $\mathrm{SINR}_{ij}(t)$ [35].

$$\tilde{\mathbf{s}}_j^{(i)}(t) = \begin{cases} \mathbf{s}_j(t), & \text{if packet arrives} \\ \text{None}, & \text{if packet lost} \end{cases} \tag{22}$$

Next, we present the model for determining packet reception success. The SINR is derived by [36]

$$\mathrm{SINR}_{ij}(t) = \frac{|H_{ij}(t)|^2 P_{\mathrm{tran},i}^{\mathrm{COMM}}}{N_0^{\mathrm{COMM}} + P_{ih}^{\mathrm{jam,\ COMM}}(t)}, \tag{23}$$

where $P_{\mathrm{tran},i}^{\mathrm{COMM}}$ is transmit power of node i, N_0^{COMM} denotes the thermal noise power spectral density over the communication bandwidth, $P_{ih}^{\mathrm{jam,\ COMM}}$ represents the jammer power received from the jammer h targeting the communication of UAV i, and H_{ij} denotes the channel power gain from UAV i to UAV j. The packet reception rate $\mathbb{P}_{\mathrm{recv},ij}(t)$ as a function of $\mathrm{SINR}_{ij}(t)$ and frame size f is modeled as [35]

$$\mathbb{P}_{\mathrm{recv},ij}(t) = \left(1 - \frac{1}{2}\exp\left(-\frac{\mathrm{SINR}_{ij}(t)}{2} \cdot \frac{B_N^{\mathrm{COMM}}}{R^{\mathrm{COMM}}}\right)\right)^{8f}, \tag{24}$$

where R^{COMM} denotes the physical-layer data rate (in bits per second) and B_N^{COMM} is the effective noise bandwidth (in Hz). For typical UAV communication links based on IEEE 802.11 protocols, we adopt $R^{\mathrm{COMM}} = 6$ Mbps and $B_N^{\mathrm{COMM}} = 20$ MHz. This formulation captures the exponential decay of bit error probability with increasing SINR, compounded over the $8f$ bits in a packet. Here, we adopt a representative control frame size of $f = 128$ bytes (i.e., 1024 bits). The resulting $\mathbb{P}_{\mathrm{recv},ij}(t) - \mathrm{SINR}_{\mathrm{ij}}(\mathrm{t})$ curve exhibits a characteristic S-shaped "cliff" behavior, as illustrated in Fig. 3. Thus, the packet loss rate is given by

$$\mathbb{P}_{\mathrm{loss},ij}(t) = 1 - \mathbb{P}_{\mathrm{recv},ij}(t). \tag{25}$$

Then, the UAV node i estimates neighbor j's state via hold-last or linear prediction, given by

$$\hat{\mathbf{s}}_j^{(i)}(t) = \begin{cases} \tilde{\mathbf{s}}_j^{(i)}(t_m), & t_m \le t < t_{m+1},\ \text{no new packet} \\ \mathbf{s}_j(t), & \text{packet received} \end{cases}, \tag{26}$$

where $\eta_{ij}(t)$ denotes the residual estimation error after decoding the received packet.

We take node $i \in \mathcal{V}_k$ and its reference node r_k as an example. In the event of packet loss during the time slot $[t_m, t_{m+1})$, node i cannot update its estimate of the state of r_k and thus holds the last successfully received value. Consequently, the communication-induced position error over this interval is approximated by the change of the estimated position of r_k between consecutive sampling instants, given by

$$\Delta\mathbf{p}_{i\to r_k}^{\mathrm{COMM}}(t_{[m,\ m+1)}) = \hat{\mathbf{p}}_{r_k}(t_m) - \hat{\mathbf{p}}_{r_k}(t_{m+1}), \tag{27}$$

where $\hat{\mathbf{p}}_{r_k}(t_m)$ is the most recent available estimate of the position of the reference node r_k at time t_m. If packet losses occur over a sequence of

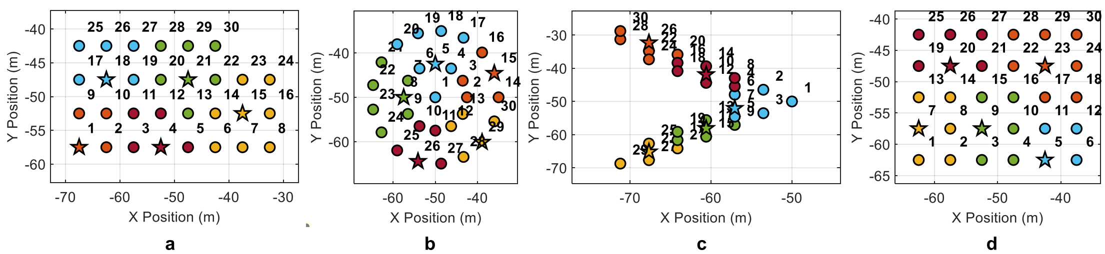

Fig. 4. Illustration of five swarm formation types. Groups within each formation are differentiated by color, and their respective reference points are indicated by stars. **a** Rectangle-like (abbreviated as "Rectangle"), **b** Circle, **c** V-shape, **d** Square-like (abbreviated as "square").

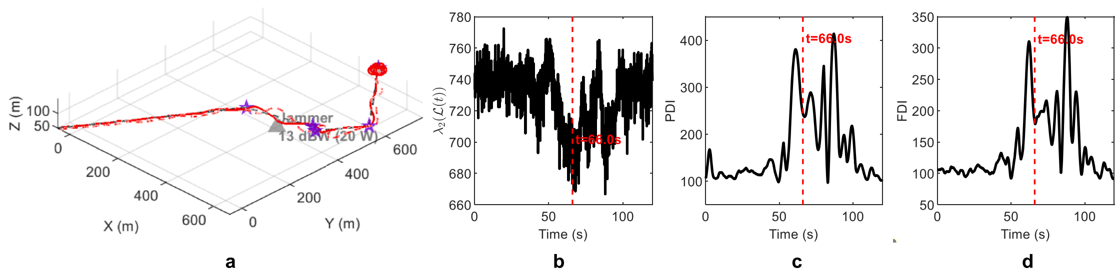

Fig. 5. Simulation results for a single trial of an 8-UAV swarm navigating through five waypoints under a 13 dBW EMI jammer targeting the GNSS of the swarm: **a** trajectories of the UAV swarm, where the black dot line represents the desired trajectory of the reference node, the red line represents the actual trajectory of the reference node, and the red dot line represents the actual trajectory of a follower node; the jammer is marked with a triangle, and the waypoints are marked with purple stars; **b** time evolution of the algebraic connectivity $\lambda_2(\mathcal{L})(t)$; **c** PDI; and **d** FDI.

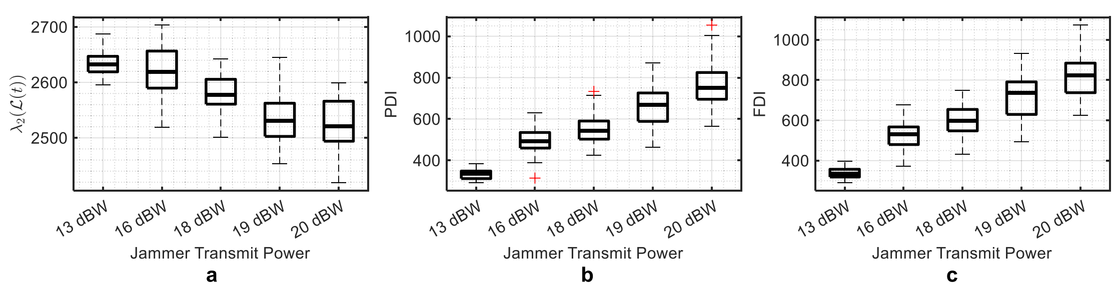

Fig. 6. Box plots summarizing simulation results from 50 Monte Carlo trials per scenario with jammers targeting the GNSS of UAV swarms under jamming power levels of {13.01 dBW, 16.02 dBW, 17.78 dBW, 19.03 dBW, 20.00 dBW}: **a** $\lambda_2(\mathcal{L}(t))$, **b** PDI, and **c** FDI.

consecutive time slots, the total accumulated communication error at time $t \in [t_s, t_{s+l})$ is given by the sum of errors over all back-to-back lost intervals up to t, expressed as

$$\Delta \mathbf{p}^{\mathrm{COMM}}_{i\to r_k}(t) = \sum_{m=s_0}^{s+l} \Delta \mathbf{p}^{\mathrm{COMM}}_{i\to r_k}([t_m, t_{m+1})), \tag{28}$$

where s_0 denotes the index of the first time slot in the current burst of consecutive packet losses.

3.4. Joint navigation-communication error propagation

The overall path and formation deviations of a UAV swarm are determined by the individual position errors of its members. For reference node r_k, the primary source of position error is navigation interference from GNSS jamming. At each time step, the controller drives the UAV toward its desired position based on the GNSS-derived estimate, i.e.,

$$\mathbf{p}^{\mathrm{des}}_{r_k}(t) = \hat{\mathbf{p}}_{r_k}(t). \tag{29}$$

However, due to the presence of jammers, the estimated position $\hat{\mathbf{p}}_{r_k}(t)$ deviates from the true position $\mathbf{p}_{r_k}(t)$, as modeled in Eq. (21). Consequently, the position error of reference node r_k is given by

$$e_{r_k}(t) = \mathbf{p}_{r_k}(t) - \mathbf{p}^{\mathrm{des}}_{r_k}(t) = -\Delta \mathbf{p}^{\mathrm{GNSS}}_{r_k}(t). \tag{30}$$

For a follower node $i \in \mathcal{V}_k$, position deviations arise from two sources: (i) GNSS navigation errors affecting their own localization, and (ii) communication-induced errors in acquiring the state of reference node r_k. The desired position of node i is defined relative to the leader, derived by

$$\mathbf{p}^{\mathrm{des}}_{i}(t) = \tilde{\mathbf{p}}_{r_k}(t) + d_{i\to r_k}, \tag{31}$$

where $d_{i\to r_k}$ is the prescribed relative offset, and $\tilde{\mathbf{p}}_{r_k}(t)$ is the position estimate of reference node r_k available to node i. This estimate is corrupted by both the GNSS error of reference node r_k and communication disruptions. Therefore, the error in the desired position of node i is

$$e^{\mathrm{des}}_{i}(t) = \Delta \mathbf{p}^{\mathrm{GNSS}}_{r_k}(t) + \Delta \mathbf{p}^{\mathrm{COMM}}_{i\to r_k}(t), \tag{32}$$

where $\Delta \mathbf{p}^{\mathrm{COMM}}_{i\to r_k}(t)$ denotes the accumulated communication-induced error in tracking the trajectory of reference node r_k. Finally, the total position error of node i combines the deviation in its desired position and its own GNSS localization error, described as

$$e_i(t) = e^{\mathrm{des}}_{i}(t) - \Delta \mathbf{p}^{\mathrm{GNSS}}_{i}(t) = \Delta \mathbf{p}^{\mathrm{GNSS}}_{r_k}(t) + \Delta \mathbf{p}^{\mathrm{COMM}}_{i\to r_k}(t) - \Delta \mathbf{p}^{\mathrm{GNSS}}_{i}(t). \tag{33}$$

4. Measurements for the degradation of the coordination of UAV swarms

In this Section, we propose three indexes which span both the control and information layers of UAV swarms. The combined performance of the three indexes reflects the overall quality of mission execution [37]. Formation distortion index captures how accurately a swarm maintains its intended geometric structure, directly indicating collective control fidelity. Path deviation index measures the swarm's ability to track desired trajectories, reflecting navigation reliability. Communication effectiveness, evaluated via packet success rate and communication degradation index, ensures robust exchange of critical data such as commands and video streams.

4.1. Communication degradation index

The successful reception of data packets among nodes in UAV swarms is inherently probabilistic. In Section 3.3, we examine how this probabilistic behavior influences the reception of state packets. Although such behavior is probabilistic, the connectivity of communication networks can effectively reflect the overall system communication performance. It should also be noted that, in addition to state exchanges, communication links must support the transmission of mission-critical information. Network connectivity governs the reachability and dissemination efficiency of information across the swarm, directly affecting consensus convergence and coordination resilience. Meanwhile, the bandwidth of inter-UAV links determines the volume and rate of data that can be transmitted, influencing the timeliness and fidelity of shared information such as state estimates, control commands, and environmental data.

The communication topology of the UAV swarm is modeled as a time-varying weighted directed graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t), \mathcal{W}(t))$, where $\mathcal{V} = \{1, 2, \dots, N\}$ denotes the set of UAVs and $\mathcal{E}(t) \subseteq \mathcal{V} \times \mathcal{V}$ represents the set of active directional communication links at time t. The SINR between two UAVs may be direction-dependent in practice. Here, we construct a symmetric weight matrix $\mathcal{W}(t) = [w_{ij}(t)] \in \mathbb{R}^{N\times N}$ by symmetrizing the directional SINR measurements. Specifically, let $\mathrm{SINR}_{ij}(t)$ and $\mathrm{SINR}_{ji}(t)$ denote the measured SINR from UAV i to j and from j to i, respectively. The edge weight of $\mathcal{G}(t)$ is then defined as

$$w_{ij}(t) = \mathrm{SINR}_{ij}(t) + \mathrm{SINR}_{ji}(t), \tag{34}$$

which ensures the symmetry $w_{ij}(t) = w_{ji}(t)$. Here, $w_{ij}(t) \geq 0$ reflects the combined quality of the bidirectional link between UAV i and UAV j. Moreover, here, we transform the $w_i j(t)$ to dB to facilitate graphical representation in the subsequent analysis. Then, we construct the associated time-varying weighted Laplacian matrix $\mathcal{L}(t) \in \mathbb{R}^{N\times N}$, defined as

$$\mathcal{L}(t) = D(t) - W(t), \tag{35}$$

where $D(t) = \mathrm{diag}(d_1(t), \dots, d_N(t))$ is the degree matrix with diagonal entries $d_i(t) = \sum_{j=1}^{N} w_{ij}(t)$. Owing to the symmetry of $\mathcal{W}(t)$, $\mathcal{L}(t)$ is symmetric positive semi-definite, and its eigenvalues are real and non-negative. Let $0 = \lambda_1(\mathcal{L}(t)) \leq \lambda_2(\mathcal{L}(t)) \leq \cdots \leq \lambda_N(\mathcal{L}(t))$ denote the ordered spectrum of $\mathcal{L}(t)$. The second smallest eigenvalue, $\lambda_2(\mathcal{L}(t))$, known as the *algebraic connectivity* or *Fiedler value*, plays a pivotal role in characterizing the connectedness of the network at time t. A larger $\lambda_2(\mathcal{L}(t))$ indicates stronger global connectivity and faster information diffusion across the swarm, while $\lambda_2(\mathcal{L}(t)) = 0$ implies that the graph is disconnected. Here, $\lambda_2(\mathcal{L}(t))$ not only reflects whether the network is connected but also captures the robustness of connectivity against node or link failures. In the context of electromagnetic interference, degradation in link quality leads to a reduction in $\lambda_2(\mathcal{L}(t))$, potentially driving the swarm toward fragmentation Table 1.

4.2. Formation distortion index

To quantify the degradation of spatial coherence in a UAV swarm under EMI, we introduce the *Formation Distortion Index* (FDI). Let $\mathbf{p}_i(t) \in \mathbb{R}^3$ denote the position of the i-th UAV at time t, and let $\mathcal{G}^{\mathrm{des}}(t) = (\mathcal{V}, \mathcal{E}(t))$ represent the desired communication or geometric topology, where $\mathcal{V}$ is the set of UAVs and $\mathcal{E}^{\mathrm{des}}(t)$ encodes the intended neighbor relationships. For each pair of connected UAVs $(i, j) \in \mathcal{E}^{\mathrm{des}}(t)$, the desired inter-agent distance is denoted by d_{ij}^{des}.

The FDI is defined as the root-mean-square (RMS) deviation of actual inter-agent distances from their nominal values, derived by

$$\mathrm{FDI} = \sqrt{\frac{1}{|\mathcal{E}^{\mathrm{des}}|} \sum_{(i,j)\in\mathcal{E}^{\mathrm{des}}} \left(\|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| - d_{ij}^{\mathrm{des}} \right)^2 }. \tag{36}$$

A value of FDI = 0 indicates perfect formation maintenance, while larger values reflect increasing structural distortion. This index captures collective coordination quality and is invariant to rigid-body translations or rotations of the entire swarm, making it suitable for mission-agnostic reliability assessment.

4.3. Path deviation index

To evaluate trajectory-level performance degradation caused by EMI-induced navigation errors, we define the *Path Deviation Index* (PDI). In Section 3.4, we define the desired position of each UAV as $\mathbf{p}^{\mathrm{des}}(t)$, which is computed based on GNSS estimates and is therefore subject to GNSS-induced errors. Assuming that each UAV i is assigned a reference trajectory over the mission horizon $[0, T]$, we define the ideal trajectory, namely the trajectory that would be followed in the absence of estimation or communication disturbances, as $\mathbf{tr}_i(t)$. The actual trajectory, denoted $\mathbf{p}_i(t)$, may deviate due to compromised positioning or communication. The PDI aggregates individual tracking errors across the swarm, given by

$$\mathrm{PDI} = \frac{1}{NT} \int_0^T \sum_{i=1}^{N} \|\mathbf{p}_i(t) - \mathbf{tr}_i(t)\| \, dt, \tag{37}$$

where N is the number of UAVs. For discrete-time implementations with sampling interval Δt and $K^{\mathrm{step}} = T/\Delta t$ steps, the PDI is approximated as

$$\mathrm{PDI} = \frac{1}{NK} \sum_{k=1}^{K^{\mathrm{step}}} \sum_{i=1}^{N} \|\mathbf{p}_i(k\Delta t) - \mathbf{tr}_i(k\Delta t)\|. \tag{38}$$

The PDI reflects the average spatial deviation per UAV over the mission duration; lower values indicate better path-following reliability. Unlike mission-specific success criteria, the PDI provides a general, quantitative measure of navigation integrity under EMI stress, supporting cross-mission resilience evaluation.

5. Numerical experiments

All simulations are conducted on a computer with an Intel(R) Core(TM) Ultra 7 155H (1.40 GHz) processor, 32 GB of RAM, using MATLAB R2023b. Experiments in this paper preserving the control and GNSS/communication models of the main study. We define five waypoints within an $800 \times 800 \times 100$ m^3 operational area, and the UAVs are required to visit them sequentially. Due to the large operational space, jammers are strategically placed within a confined circular region (50-meter radius) centered at the midpoint of the UAV trajectory, with a vertical distribution ranging from 0 to 200 m. Unless otherwise stated, one jammer is deployed per simulation. The jammers targeting GNSS are configured at a power level of 16.02 dBW, and the jammers targeting communication networks operate at 25.12 dBW. Figs. 4**a**, 4**b**, 4**c**, and 4**d** illustrates the four formation types considered in this study, including rectangle-like formation (denoted as "Rectangle"), circular formation, V-shaped formation, and square-like formation (denoted as "Square"). Unless otherwise specified, the rectangle-like formation is adopted throughout the simulations.

Fig. 5 shows the time-varying performance of a UAV swarm in a trial with a jammer directed at GNSS set to 16.02 dBW. Fig. 5**a** show the trajectories of the UAV swarm. Figs. 5**b**, 5**c**, and 5**d** depict the time evolution of the PDI, FDI, and $\lambda_2(\mathcal{L}(t))$, respectively, over the simulation period. The UAV swarm reaches its closest proximity to the jammer at $t = 66$ s, marked by a vertical dashed line in Figs. 5**b**, 5**c** and 5**d**. Around

Table 1
Model and scenario parameters for the numerical experiments.

Category	Parameter	Value	Description
Swarm Setup	N	30	Number of UAVs
	Δt	0.1 s	Time step
	T_{sim}	120 s	Total simulation time
	N_{leaders}	5	Number of leaders (6 UAVs/leader)
Control Parameters	k_p	2.0	Position gain
	k_d	2.0	Velocity gain
	$u_{\max}$	$8.0\,\mathrm{m/s}^2$	Maximum control input (acceleration limit)
Communication Parameters	$P^{\mathrm{COMM}}_{\mathrm{tran},i}$	20 dBW (100 W)	Transmit power of UAV nodes
	$P^{\mathrm{jam,\ COMM}}_{h}$	25.12 dBW (325 W)	Communication jammer transmit power (2.4 GHz)
	N^{COMM}_{0}	-204 dBW/Hz	Noise power spectral density
GNSS & Interference	C^{GNSS}	-130 dBm	GNSS signal power
	GDOP	4.0	Geometric dilution of precision
	$b(t)$	1×10^{-4}	Bias term
	B^{GNSS}_{h}	2 MHz	Jammer bandwidth
	$P^{\mathrm{jam,\ GNSS}}_{h}$	16.02 dBW (40 W)	GNSS jammer transmit power (L1 band)
	N^{GNSS}_{0}	-204 dBW/Hz	Noise power spectral density

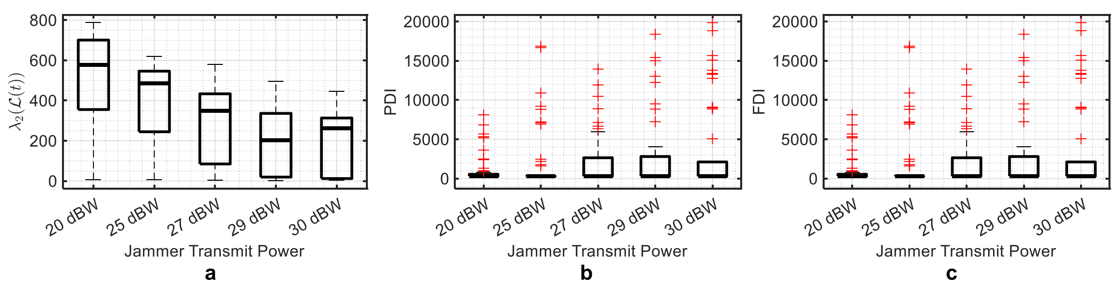

Fig. 7. Box plots summarizing simulation results from 50 Monte Carlo trials per scenario with jammers targeting communication networks of UAV swarms under jamming power levels of {20.00 dBW, 25.12 dBW, 27.40 dBW, 28.90 dBW, 30.00 dBW}: **a** $\lambda_2(\mathcal{L}(t))$, **b** PDI, and **c** FDI.

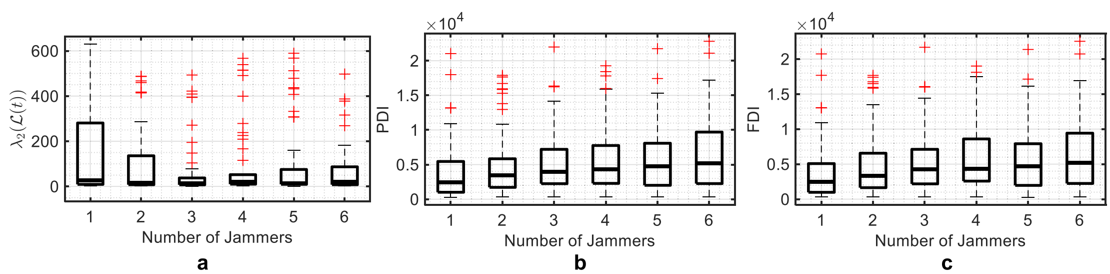

Fig. 8. Box plots summarizing simulation results from 50 Monte Carlo trials per scenario. The total jamming power directed at GNSS is 16.02 dBW, and that directed at communication networks is 25.12 dBW. The number of jammers is set to {1, 2, 3, 4, 5, 6}: **a** $\lambda_2(\mathcal{L}(t))$, **b** PDI, and **c** FDI.

this moment, both the PDI and the FDI exhibit multiple peaks, while the algebraic connectivity $\lambda_2(\mathcal{L}(t))$ shows a trough. Moreover, we observe that abrupt variations in both PDI and FDI are closely associated with directional changes of formations during waypoint transitions, where rapid changes in velocity and orientation temporarily degrade tracking performance and formation keeping accuracy.

To investigate the impact of jammer power on UAV swarm performance, we independently analyze two interference scenarios: GNSS jamming and communication jamming. Here, we employ a single jammer with its location randomly varied across trials. We conduct 50 Monte Carlo simulations. First, for GNSS jamming scenarios, the transmit power of GNSS jammer is varied from 13.01 dBW to 20.00 dBW. Here, the transmit power of communication jammer is set to 0 W. The results are presented in Figs. 6**a**, 6**b**, and 6**c**. Then, for communication jamming scenarios, the transmit power is varied from 20.00 dBW to 30.00 dBW, with results presented in Figs. 7**a**, 7**b**, and 7**c**. Here, the transmit power of GNSS jammer is set to 0 W. These results substantiate the claim that stronger electromagnetic interference worsens tracking accuracy and formation coherence by impeding command dissemination and GNSS accuracy. Moreover, the comparison of Fig. 6 and Fig. 7 demonstrate that GNSS jammers have a more pronounced effect on both PDI and FDI compared to communication jammers.

Then, under a fixed total jamming power budget, we compare the resulting coordination degradation between a single high-power jammer and multiple spatially distributed low-power jammers with equivalent aggregate power to assess how the spatial concentration versus dispersion of interference affects swarm-wide connectivity and formation stability. Here, we fix the total jamming power directed at GNSS and communication systems at 16.02 dBW and 25.12 dBW, respectively. This total power is evenly distributed among all co-located jammers. In all simulations, the GNSS and communication jammers are co-located at the same randomly chosen position, which is uniformly sampled within a horizontal range of ± 100 m from the trajectory midpoint and a vertical range of 0 to 200 m. A total of 50 independent Monte Carlo trials are conducted. The results are shown in Figs. 8**a**, 8**b**, and 8**c**. As shown, both the PDI and FDI exhibit a slight upward trend as the number of jammers increases. Communication connectivity, quantified by algebraic connectivity $\lambda_2(\mathcal{L}(t))$, exhibits a decreasing trend as the number of jammer in-

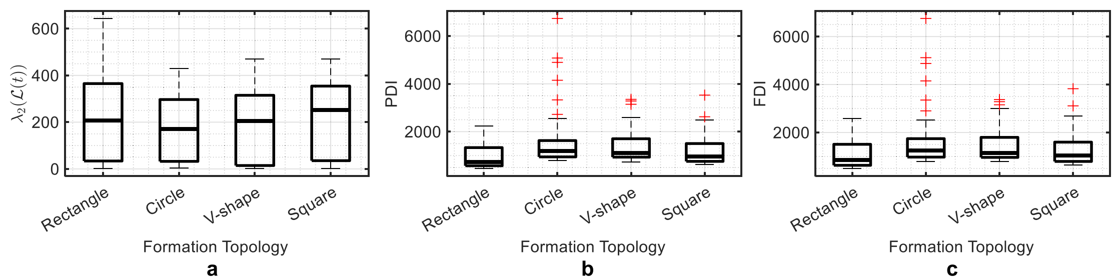

Fig. 9. Box plots summarizing the simulation results of 50 Monte Carlo trials across four UAV swarm formations, Rectangle-like, Circle, V-shape, and Square-like: **a** $\lambda_2(\mathcal{L}(t))$, **b** PDI, and **c** FDI. The total jamming power directed at GNSS is 16.02 dBW, and that directed at communication networks is 25.12 dBW.

creases in the scenarios considered. Overall, the results highlight that, in addition to interference intensity, spatial coverage also plays a critical role in degrading the coordination performance of UAV swarms.

From the standpoint of improving the resilience of UAV swarms under EMI, four canonical formation geometries illustrated in Fig. 4 are evaluated. In this scenario, we use one jammer. The jammer locations are randomly varied across 50 independent Monte Carlo trials. The results are shown in Figs. 9**a**, 9**b**, and 9**c**. The rectangle-like and square-like formations both demonstrate strong performance across the three indexes. However, while the square-like formation exhibits superior communication connectivity, the rectangle-like formation achieves better performance in terms of PDI and FDI.

6. Conclusion

In this study, we develop a physics-informed analytical framework to systematically characterize the impact of EMI on the coordination performance of multi-agent UAV swarms. More specifically, we quantitatively demonstrate that increasing jamming power leads to higher packet loss rates in inter-UAV communication links, which in turn disrupts the reliable dissemination of control commands and the timely exchange of state information necessary for maintaining formation geometry and synchronized motion. At the same time, we formalize the degradation of GNSS-based positioning accuracy under EMI by establishing a physically grounded relationship between jamming intensity and navigation error, capturing both magnitude and statistical characteristics of the induced position uncertainty. Moreover, we propose a set of indicators that effectively capture key aspects of UAV swarm coordination, including path following accuracy, formation maintenance, and communication integrity. Experimental results demonstrate that distributed jammer deployment can significantly degrade swarm coordinated behavior.

For future work, it would be valuable to investigate more sophisticated and intelligent jamming patterns. Additionally, more safety-related questions can be examined, such as whether EMI can lead to collisions among UAVs and, if so, whether such collisions could propagate through the swarm, triggering cascading failures.

CRediT authorship contribution statement

Meixuan Jade Li: Writing – original draft, Investigation, Conceptualization, Formal analysis, Software, Validation, Visualization; **Cheng Zhu:** Writing – review & editing, Resources; **Xianqiang Zhu:** Writing – review & editing, Resources; **Mingrui Lao:** Writing – review & editing, Resources; **Hanning Tang:** Writing – review & editing, Investigation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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